

Nathan Jacobson

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WORK EXPERIENCE

Tae Kwon Do Instructor

September 2025 — December 2025

Transcending Martial Arts

Port Moody, BC

- Instructed children aged 4-14 martial arts techniques such as strikes, blocks, patterns, and weapons training
- Led a group of volunteer assistant instructors to aid with class management
- Developed structured lesson plans based on the class' age and skill levels

Information Officer

April 2025

Elections Canada

Port Moody, BC

- Confirmed voter identification when they entered the polling station
- Directed voters to the correct stations and assisted voters with physical disabilities
- Recorded vote counts after polls had closed

VOLUNTEER EXPERIENCE

Assistant Tae Kwon Do Instructor

June 2022 — August 2023

Transcending Martial Arts

Port Moody, BC

- Aided instructors in teaching children aged 4-9 martial arts techniques
- Managed the class, ensuring students were focused and helped those who were struggling with the lesson

Summer Camp Volunteer

July 2023

Simon Fraser University Summer Camps

Burnaby, BC

- Assisted instructors in running chemistry and engineering camps for children ages 5-6
- Supervised groups of children during structured activities, lunch breaks, and free time

PROJECTS

interchange (github.com/grqphical/interchange)

August 2025 — December 2025

- Created a reverse proxy in Go that supports proxying requests, static file hosting, HTTPS, and IP blacklisting/whitelisting

webc (github.com/grqphical/interchange)

February 2026 — Present

- Currently building a C compiler written in Go that compiles C99 code and outputs WebAssembly

Catan AI (github.com/grqphical/catan-ai)

January 2026 — Present

- Currently Building a web app with React, Python, and FastAPI that plays the board game Settlers of Catan
- Players input their board setup in the web app, update the game state each turn. It then will determine the best move that turn

cgraph (github.com/grqphical/cgraph)

December 2025 — Present

- Created a C (C99) library that can generate simple graphs and plots
- Only requires a single header and has no dependencies; the entire rendering stack is written from scratch

SKILLS

- **Programming Languages:** Python, Go, C, C++, HTML/CSS/JS, SQL
- **Libraries/Tools:** Linux, numpy, pandas, FastAPI, React, SQLite, matplotlib, Docker
- **Non-Programming Skills:** Image Editing, MS Office, Communication, Ability to work autonomously and in groups

EDUCATION

Simon Fraser University

Burnaby, BC

Bachelor's of Science, Computing Science

September 2025 — Present

- Relevant course work: Data Structures and Algorithms, Systems Programming, Linear Algebra, and Discrete Mathematics

Heritage Woods Secondary

Port Moody, BC

BC Dogwood Dipolma (High School Diploma)

September 2021 — June 2025

- Awarded Top Producing Grade 12 Student and BC Achievement Scholarship
- Graduated with Honour Roll with Distinction

EXTRACURRICULARS

Tae Kwon Do

September 2017 — Present

- Achieved a 1st Degree Black Belt